

Preclass Learning: Missing Data (ICA 8)

1. Mechanisms of Missingness

The cause of missingness dictates the appropriate handling strategy:

- **Missing Completely At Random (MCAR):** Missingness is unrelated to any variable (observed or unobserved).
- **Missing At Random (MAR):** Missingness is related to **observed** data (e.g., older subjects are less likely to answer a survey question, but we observe their age).
- **Missing Not At Random (MNAR):** Missingness is related to the **unobserved** value itself (e.g., people with low income refuse to state their income).

2. Diagnosis and Assessment

- **Frequency:** Calculate the percentage of NAs per variable.
- **Visualization:** Use correlation plots and other tools (e.g., naniar package) to visualize the *patterns* of missingness to identify if they are clustered or random.

3. Handling Strategies

Method	Description	Consideration
Listwise Deletion	Removing any row that contains at least one missing value.	Only unbiased if data is MCAR; loses significant sample size.
Mean/Median Imputation	Replacing NAs with the mean or median of the variable.	Artificially reduces variability and can distort distributions.
Model-Based Imputation	Using a statistical model (e.g., regression or tree-based methods) to predict the missing value based on other variables.	Assumes MAR; generally more accurate than simple imputation.